

C Programs for Problems 6 to 10

6. Transpose of a Matrix

```
#include <stdio.h>

int main() {

    int r, c, matrix[10][10], transpose[10][10];

    printf("Enter the number of rows and columns: ");

    scanf("%d %d", &r, &c);

    printf("Enter the elements of the matrix:\n");

    for (int i = 0; i < r; i++) {

        for (int j = 0; j < c; j++) {

            scanf("%d", &matrix[i][j]);

        }

    }

    for (int i = 0; i < r; i++) {

        for (int j = 0; j < c; j++) {

            transpose[j][i] = matrix[i][j];

        }

    }

    printf("Transpose of the matrix:\n");

    for (int i = 0; i < c; i++) {

        for (int j = 0; j < r; j++) {

            printf("%d ", transpose[i][j]);

        }

        printf("\n");

    }

}
```

```
    return 0;
}
```

7. Calculate Power Using Recursion

```
#include <stdio.h>

int power(int base, int exp) {
    if (exp == 0)
        return 1;
    return base * power(base, exp - 1);
}

int main() {
    int base, exp;

    printf("Enter base and exponent: ");

    scanf("%d %d", &base, &exp);

    printf("%d^%d = %d", base, exp, power(base, exp));

    return 0;
}
```

8. String Reversal without Built-in Function

```
#include <stdio.h>

void reverseString(char str[]) {
    int length = 0, i;

    char temp;

    while (str[length] != '\0')
```

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        length++;

    for (i = 0; i < length / 2; i++) {

        temp = str[i];

        str[i] = str[length - i - 1];

        str[length - i - 1] = temp;

    }

}

int main() {

    char str[100];

    printf("Enter a string: ");

    gets(str);

    reverseString(str);

    printf("Reversed string: %s", str);

    return 0;

}

```

9. Calculate GCD and LCM

```

#include <stdio.h>

int gcd(int a, int b) {

    if (b == 0)

        return a;

    return gcd(b, a % b);

}

int main() {

    int a, b, gcdValue, lcmValue;

```

```

printf("Enter two numbers: ");

scanf("%d %d", &a, &b);

gcdValue = gcd(a, b);

lcmValue = (a * b) / gcdValue;

printf("GCD: %d\nLCM: %d", gcdValue, lcmValue);

return 0;

}

```

10. Remove Duplicates from an Array

```

#include <stdio.h>

int main() {

    int n, arr[100], unique[100], count = 0, isDuplicate;

    printf("Enter the size of the array: ");

    scanf("%d", &n);

    printf("Enter the elements of the array: ");

    for (int i = 0; i < n; i++) {

        scanf("%d", &arr[i]);

    }

    for (int i = 0; i < n; i++) {

        isDuplicate = 0;

        for (int j = 0; j < count; j++) {

            if (arr[i] == unique[j]) {

                isDuplicate = 1;

                break;

            }

        }

    }
}

```

```
    }

    if (!isDuplicate) {

        unique[count++] = arr[i];

    }

}

printf("Array after removing duplicates: ");

for (int i = 0; i < count; i++) {

    printf("%d ", unique[i]);

}

return 0;

}
```